Hackers fight it out at UCSB to win global bragging rights

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By THOMAS SCHULTZ NEWS-PRESS STAFF WRITER

Team from 0ld Eur0pe able to 'capture the flag'

Blending cyber know-how with fighting spirit, 22 student teams from 18 universities on four continents competed Friday in a UCSB-organized confrontation billed as the largest "capture the flag" computer hacking competition ever attempted.

The goal: score the most points by defending the security of a Web site while attacking those of opponents.

The payoff: global bragging rights. And an A-plus grade for the term in some cases.

"Particularly on the defense, it's very interesting to see what real-time attacks look like," said Phil Wittrock, 23, a UCSB computer science master's degree candidate and one of 400 participants in the fray. "It's one thing to just get an assignment. It's another thing to have a running program that's vulnerable. It's much more intense. You've really got to work together."

The seven-hour "iCTF" battle started at 9 a.m. West Coast time, with UCSB faculty and research assistants manning terminals inside a makeshift control room in a first-floor lobby of Engineering I. A digital projector beamed out an automated scoreboard that continually updated.

In two adjacent rooms, two UCSB teams of 15 students talked strategy and typed -- and typed and typed and typed. By midday, two Italian teams were on top while a group of Austrians, and likewise some Germans, were coming on strong.

"The Europeans are very much controlling the game," said professor Giovanni Vigna, a key planner of the contest who teaches computer security courses at the UCSB Department of Computer Science.

A squad from the University of La Plata in Argentina represented South America. In Australia, it was Queensland University of Technology. Ten schools, including the menacing Naval Postgraduate School at Monterey, hailed from the U.S.

In the end, the Germans forged ahead and a team called 0ld Eur0pe (insisting the o's appear as zeros) took top honors. 0ld Eur0pe represented RWTH Aachen University, located near DŸsseldorf on the Germany-Belgium border.

The competition dates to 2001, when two UCSB teams battled. Other American universities joined in 2003, and last year the bout went international, said Mr. Vigna, who is known worldwide for his hacking prowess.

In July, Mr. Vigna led an elite UCSB team to a Las Vegas victory in the high-profile, three-day capture the flag competition held annually at DEFCON, the largest underground hacker convention. The DEFCON competition is widely regarded as the world championship of hacking.

Friday's contest centered on an offline Web site designed solely for the event. Each team received identical versions, which contained a variety of services typical to e-commerce. One section, for example, enabled a would-be visitor to make a financial contribution after entering personal information, while another featured a search engine.

The key strategy for all teams was to identify undisclosed problems built into the site and fix them, while leveraging that knowledge to take out competitors.

"I teach courses in information and computer security at the University of South Florida," Jeremy Rasmussen said via e-mail Friday as the competition neared its end. "There's just so much you can do in the classroom setting. We cover a lot of cryptography and network security topics, but for sake of breadth of knowledge, we can't get in-depth too much. This type of activity is invaluable for hands-on reinforcement of the concepts we teach in class.

"I don't teach hacking," he wrote. "Primarily we concentrate on defenses. Therefore, to understand the mentality of hacking, you have to put yourself in the other guy's shoes for a while. If there's anything I've learned from participating in today's competition, it's that we have some very talented students, and I need to stay sharp on my programming skills."

e-mail: tschultz@newspress.com